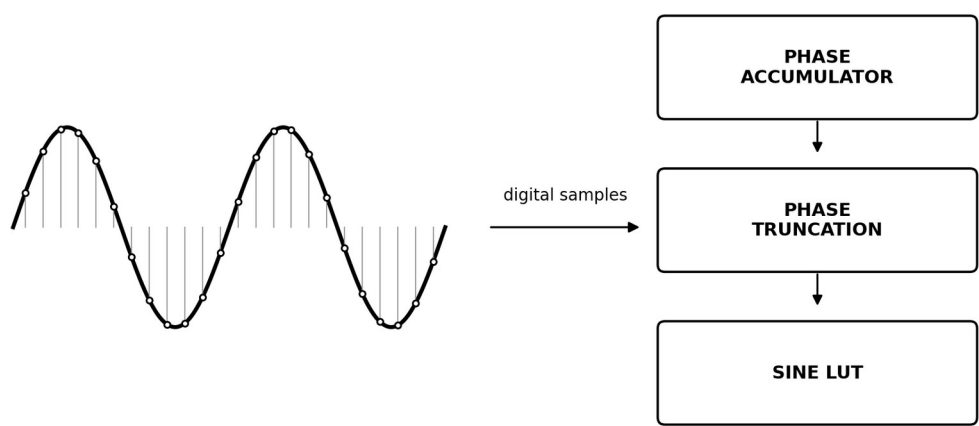


DIRECT DIGITAL SYNTHESIS

Design and operation of a DDS based on a phase accumulator and a sine lookup
table



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Package note. MATLAB source files use the .m extension. A .mat file is also supplied for numerical data and reproducibility.

1. Introduction and objectives

A direct digital synthesizer creates periodic and swept-frequency signals by converting a numerical phase into amplitude samples.

1.1 Purpose of the practice

The objective is to implement a non-recursive DDS at 48 kHz, study the effect of the phase accumulator and lookup-table dimensions, and verify the generation of three fixed tones and a repetitive linear-frequency-modulated signal.

- Build a 2048-sample full-cycle sine lookup table using Q1.31 amplitude quantization.
- Use a 32-bit phase accumulator and the 11 most significant phase bits as the LUT address.
- Generate 400 Hz, 600 Hz and 1 kHz tones and inspect their time-domain and spectral behaviour.
- Generate a 500 Hz to 3 kHz linear sweep, repeated once per second over a two-second record.
- Provide modular MATLAB code, a self-contained alternative, reference plots and reproducible data.

1.2 Configuration summary

Parameter	Value	Function
Sampling frequency	48 kHz	Sets the output sampling rate and Nyquist limit.
Accumulator width	32 bits	Defines phase resolution and tuning-word range.
LUT address width	11 bits	Selects 2048 stored sine samples.
Amplitude format	Q1.31	Represents normalized signed sine values.
Tone duration	1 s	Provides 1 Hz DFT-bin spacing for the tone tests.
LFM duration	2 s	Contains two one-second rising sweeps.

Validation note. The reference figures in this report were produced with an equivalent numerical model using the same accumulator arithmetic, LUT addressing and quantization. MATLAB was not available in the authoring environment, so the delivered .m files were checked by inspection and by numerical equivalence rather than executed directly.

2. DDS operating principle

A DDS separates frequency control from waveform storage. At every sampling instant, the phase accumulator adds an integer tuning word M . Overflow implements modulo- 2^B phase wrapping, so the accumulator naturally represents one complete cycle.

Direct Digital Synthesizer architecture

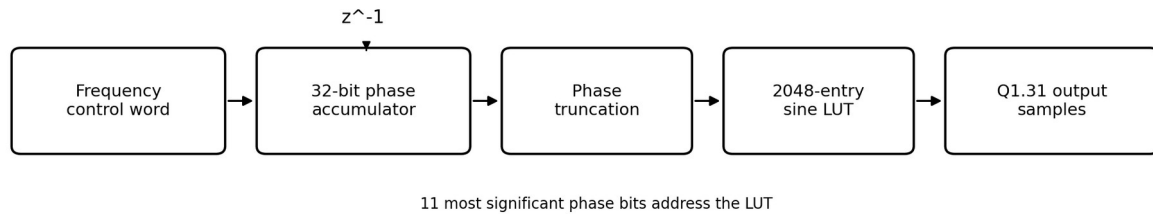


Figure 1. Functional architecture of the implemented direct digital synthesizer.

2.1 Frequency control

For an accumulator of B bits and sampling frequency f_s , the ideal tuning word for a requested output frequency f_{out} is:

$$M = \text{round}(f_{out} * 2^B / f_s)$$

The generated frequency is determined by the integer tuning word, not by the requested real-valued frequency:

$$f_{actual} = M * f_s / 2^B$$

2.2 Phase-to-amplitude conversion

Only the 11 most significant accumulator bits address the sine table. This is equivalent to discarding 21 lower phase bits. The operation reduces memory from 2^{32} possible phase states to 2048 stored amplitudes while maintaining extremely fine frequency resolution.

3. Numerical design and quantization

3.1 Frequency resolution

The smallest change in output frequency caused by increasing the tuning word by one least-significant bit is:

$$\Delta f = f_s / 2^B = 48000 / 2^{32} = 11.175870895 \text{ microhertz}$$

This resolution is much smaller than the frequency errors produced by common analogue oscillators. The remaining errors in this practice are therefore dominated by LUT phase truncation and amplitude quantization rather than by frequency-word granularity.

3.2 Sine lookup table

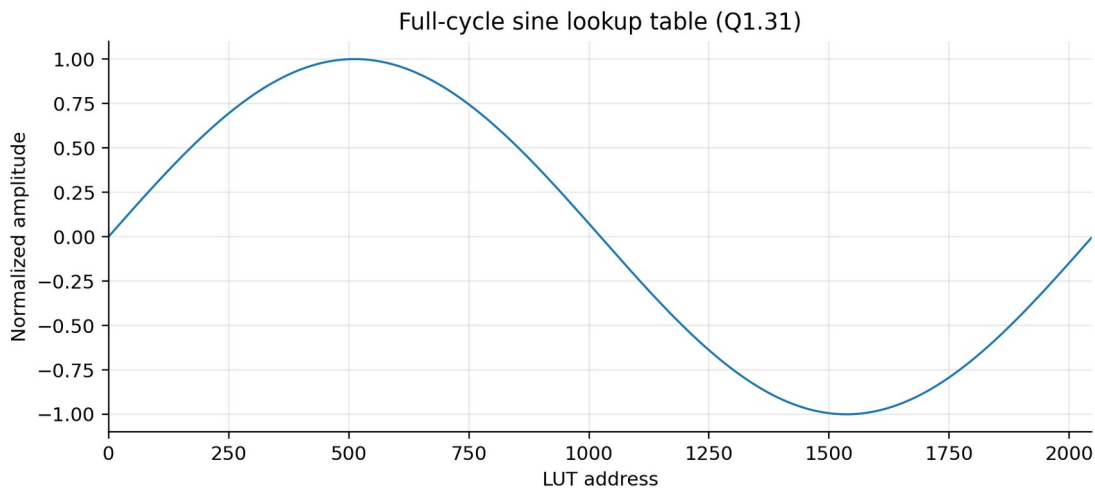


Figure 2. The 2048-entry, full-cycle sine lookup table after Q1.31 normalization.

The code first calculates uniformly spaced angles from 0 to 2π , excluding the duplicate endpoint. Each sine value is scaled to a signed 32-bit integer and converted back to a normalized floating-point sample for convenient simulation. The integer table is retained in the reference MAT file.

3.3 Expected non-idealities

- Phase truncation creates deterministic spurious components because many accumulator states map to the same LUT address.
- Q1.31 amplitude quantization introduces very small sample errors; its effect is negligible compared with the 11-bit phase address in this configuration.
- The output is sampled at 48 kHz, so all requested frequencies must remain below the 24 kHz Nyquist frequency.

4. MATLAB implementation

The recommended solution is modular. The main script defines the experiment, while four functions implement LUT construction, fixed-tone generation, LFM generation and a toolbox-independent short-time Fourier transform.

File	Responsibility
DDS_main.m	Configures the practice, creates figures and saves DDS_reference_results.mat.
dds_build_lut.m	Constructs the 2048-entry full-cycle Q1.31 sine table.
dds_generate_tone.m	Generates one fixed-frequency DDS output and reports the integer tuning word.
dds_generate_lfm.m	Generates the repeated linear frequency law and corresponding DDS samples.
dds_simple_spectrogram.m	Computes an STFT without the Signal Processing Toolbox.
DDS_single_script.m	Self-contained alternative that follows the original two-part script structure.

4.1 Core accumulator loop

```
phase_accumulator = mod(phase_accumulator + tuning_word, ...
    phase_modulus);
lut_index_zero_based = bitshift(phase_accumulator, -shift);
lut_index_matlab = double(lut_index_zero_based) + 1;
x(n) = lut(lut_index_matlab);
```

The right shift extracts the most significant phase bits. Adding one converts the zero-based hardware-style address into MATLAB indexing. The accumulator and tuning words use uint64 storage so the 32-bit unsigned arithmetic is represented exactly.

4.2 Changes with respect to the supplied draft

- All three requested tones are generated instead of only the 400 Hz case.
- The tuning word is rounded to an integer to model realizable digital hardware.
- Spectra are displayed as normalized decibels and restricted to the positive-frequency half.
- The LFM frequency law is expressed directly in hertz, then converted sample by sample into tuning words.

5. Fixed-tone results

5.1 400 Hz tone

The 400 Hz output completes one period every 2.5 ms. The time plot shows eight cycles, and the spectrum contains a dominant line at the requested frequency. Small deterministic spurs are expected from the finite phase-address width.

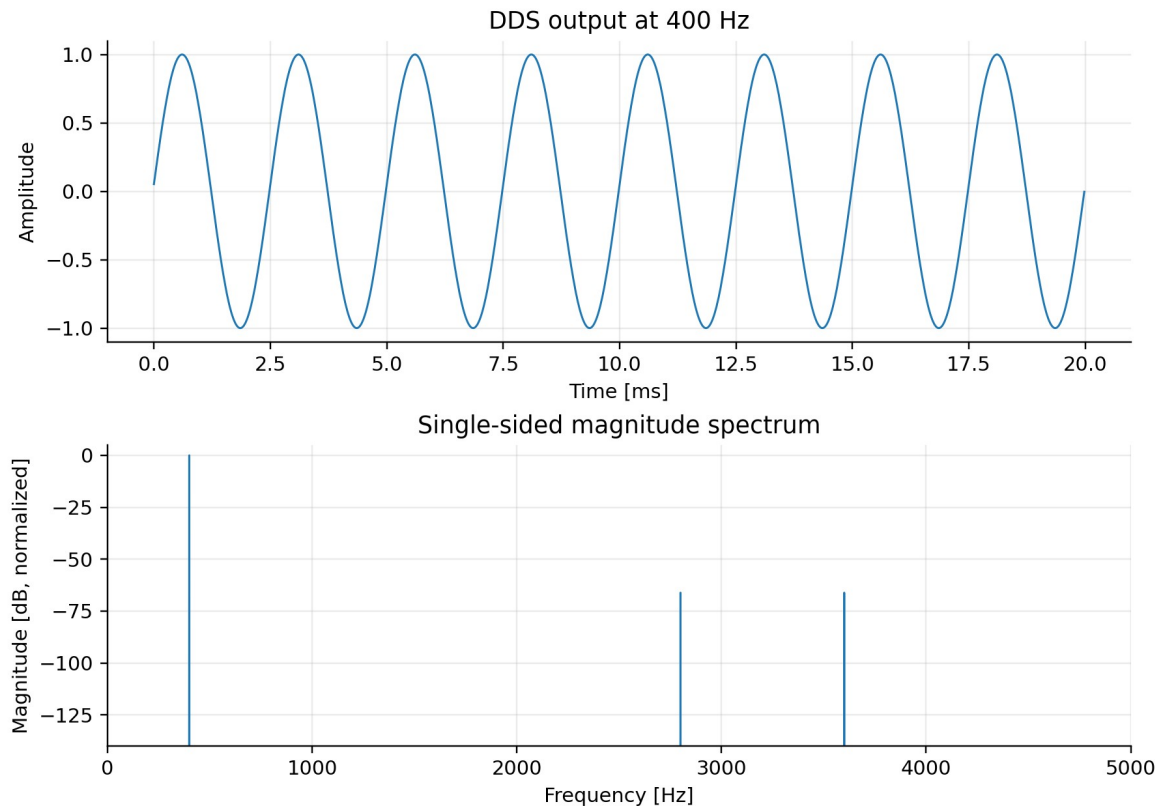


Figure 3. Time-domain waveform and single-sided spectrum for the 400 Hz DDS output.

Interpretation

- Requested tuning word: $M = 35,791,394$.
- Actual output frequency: 399.999998509884 Hz.
- Frequency error: -1.4901 microhertz, well below one DFT bin for a one-second record.

5.2. 600 Hz and 1 kHz tones

Increasing M makes the phase accumulator advance faster. The waveform period decreases from approximately 1.667 ms at 600 Hz to 1 ms at 1 kHz, while the output amplitude remains limited by the same Q1.31 sine table.

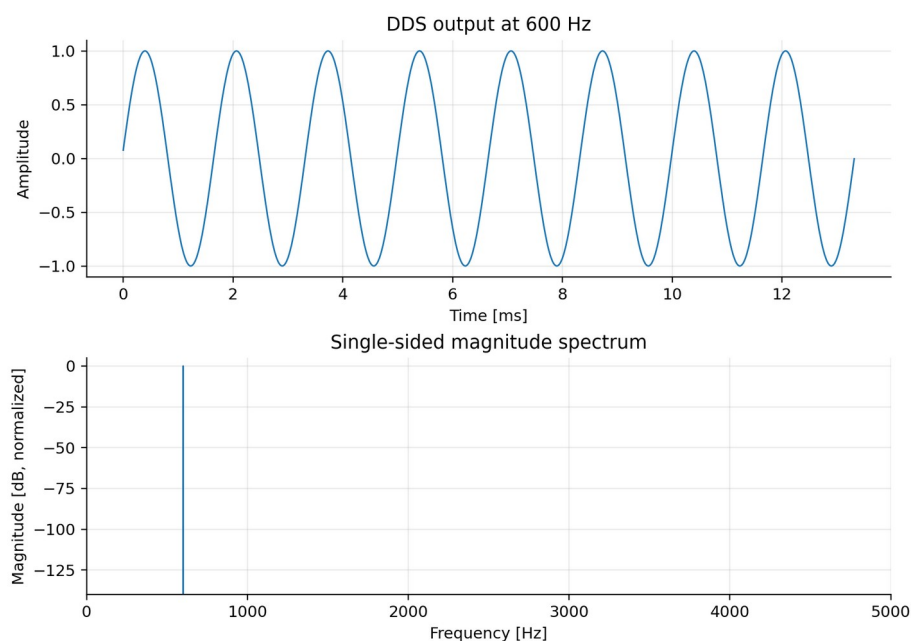


Figure 4. DDS time waveform and spectrum at 600 Hz.

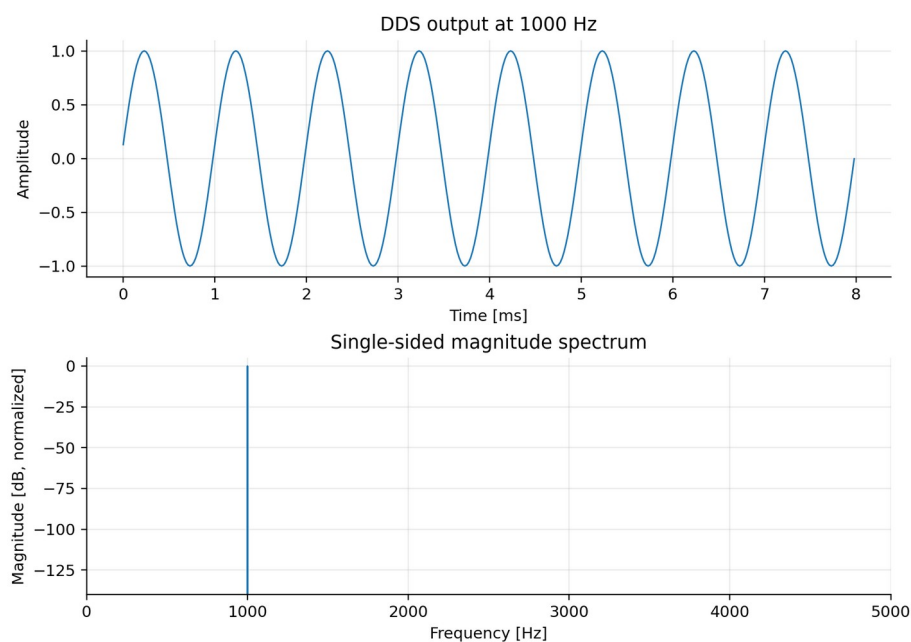


Figure 5. DDS time waveform and spectrum at 1 kHz.

5.3. Comparison and numerical accuracy

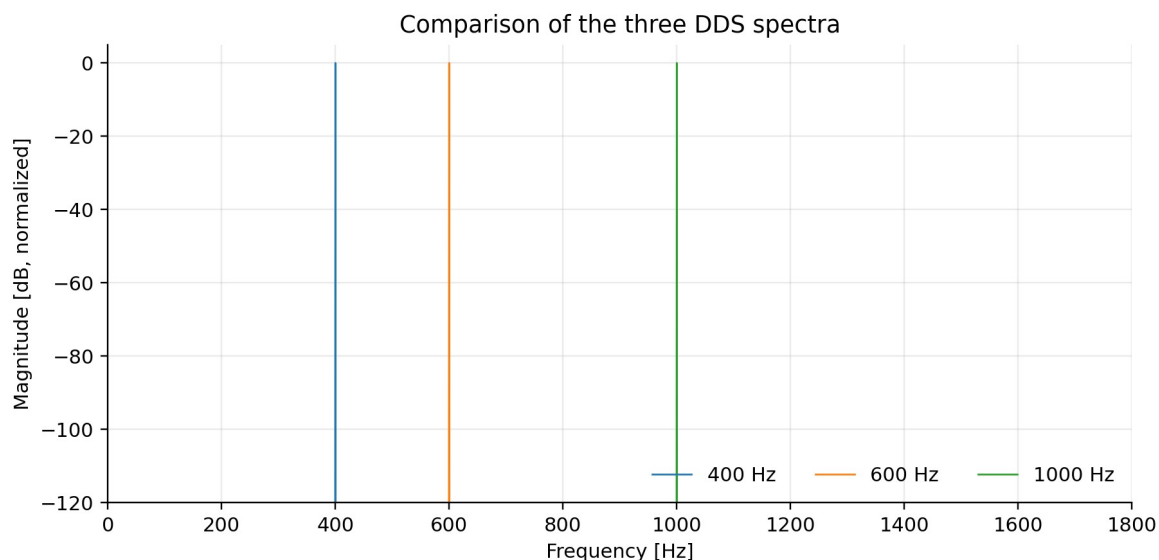


Figure 6. Overlay of the normalized spectra for the three fixed tones.

The peaks occur at their intended positions. Because the record contains exactly one second, the DFT spacing is 1 Hz. The tuning-word errors are several orders of magnitude smaller and cannot be resolved by the plotted FFT.

Requested [Hz]	Tuning word M	Actual [Hz]	Error [microHz]
400	35,791,394	399.999998509884	-1.4901
600	53,687,091	599.999997764826	-2.2352
1000	89,478,485	999.999996274710	-3.7253

Spectral interpretation

The normalized magnitude spectrum is useful for locating spurious components, but the exact spur levels depend on coherent sampling, record length and the phase-truncation pattern. A window can be applied when testing non-coherent frequencies; it is not necessary here because the requested tones are integer multiples of the 1 Hz bin spacing.

6. Linear frequency modulation

The LFM signal uses a tuning word that rises linearly from the value corresponding to 500 Hz to the value corresponding to 3 kHz. The ramp duration is one second and the complete record is two seconds, so the frequency law repeats twice.

```
sweep_position = mod(0:num_samples-1, samples_per_sweep);
f_command = f_start + (f_end - f_start) .* ...
    sweep_position / (samples_per_sweep - 1);
tuning_words = uint64(round(f_command * 2^accumulator_bits / fs));
```

For every sample, the current tuning word is added to the same phase accumulator used by the fixed-tone generator. The phase remains continuous at the ramp reset, although the instantaneous frequency jumps from 3 kHz back to 500 Hz.

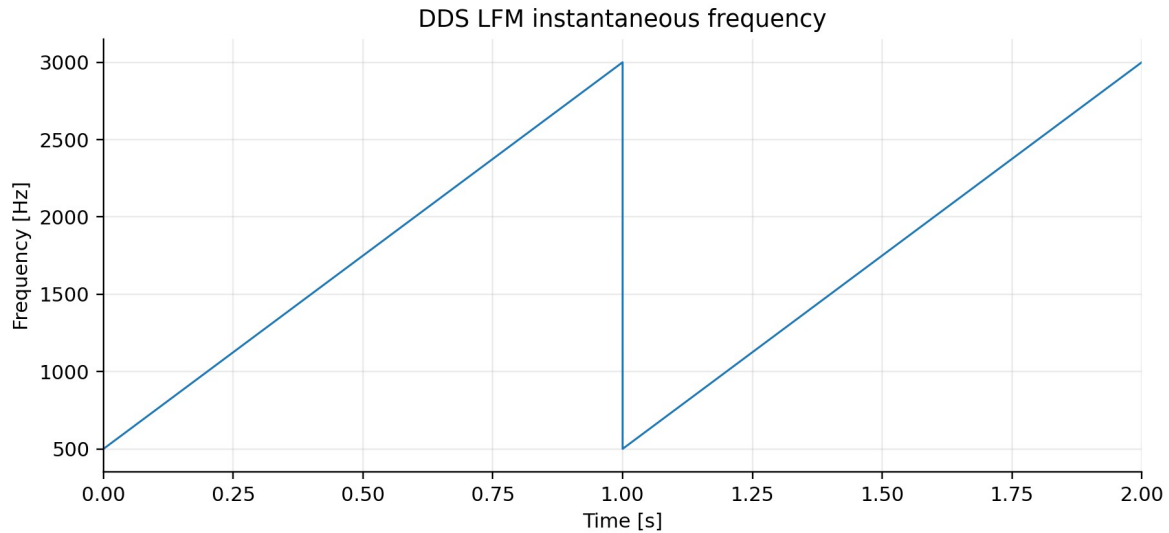


Figure 7. Commanded and quantized instantaneous-frequency law for two repeated LFM sweeps.

6.1 Sweep rate

$$\text{sweep rate} = (3000 - 500) / 1 \text{ s} = 2500 \text{ Hz/s}$$

The DDS can update the frequency once per sample, so the control law contains 48,000 frequency steps during each sweep. Integer tuning-word quantization changes each step by less than half of the 11.18 microhertz frequency resolution.

6.1. LFM spectrum and spectrogram

A conventional FFT collapses the complete two-second record into one frequency-domain view. The energy is spread across the swept band, whereas the spectrogram preserves the time evolution and reveals two rising diagonal traces.

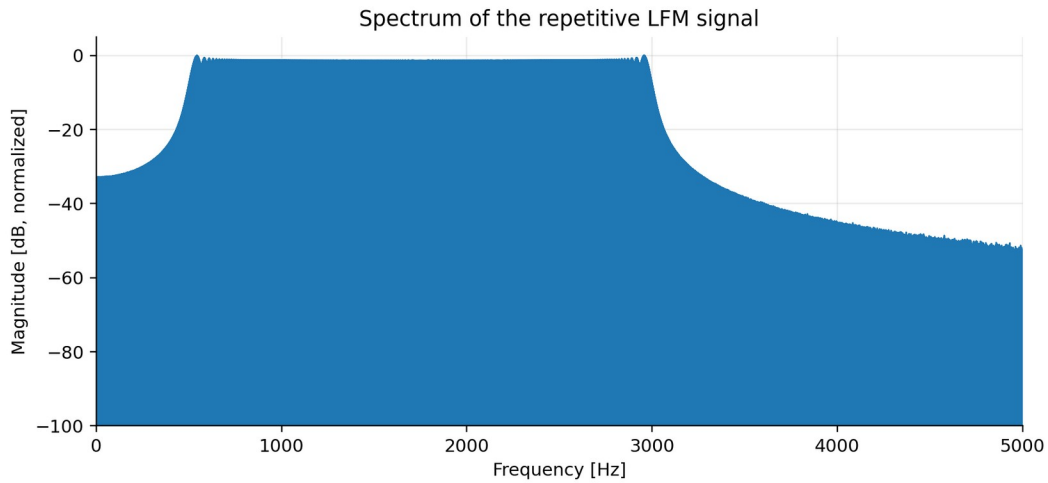


Figure 8. Whole-record spectrum of the repetitive 500 Hz to 3 kHz LFM output.

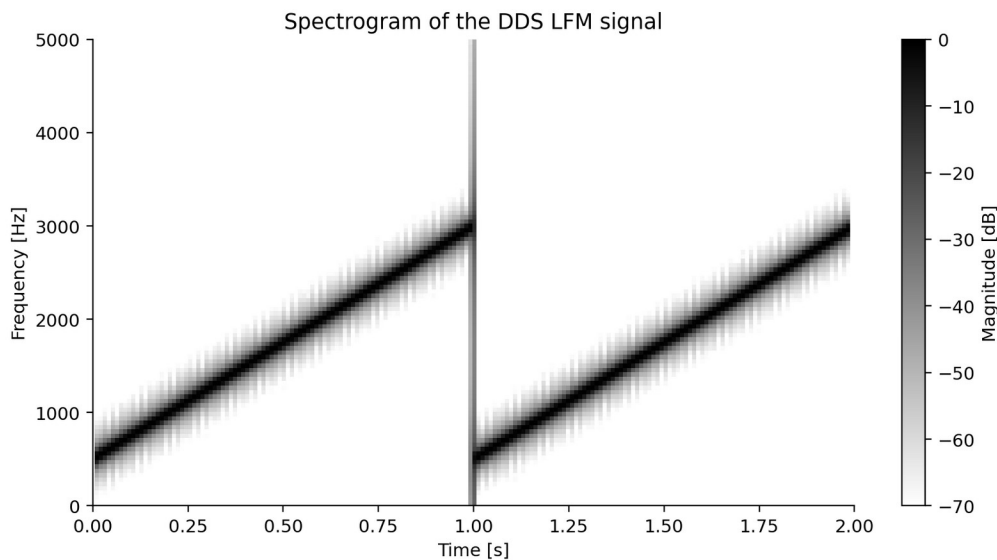


Figure 9. STFT magnitude showing the two one-second linear frequency ramps.

The STFT uses a 1024-sample Hann window, 512-sample overlap and 1024-point FFT. At 48 kHz this corresponds to a window duration of approximately 21.3 ms and a frequency-bin spacing of 46.875 Hz. The displayed ridge therefore has finite thickness even though the commanded frequency is precisely defined.

7. Discussion, limitations and validation

7.1 Design trade-offs

Design choice	Benefit	Cost or limitation
32-bit accumulator	Extremely fine frequency resolution.	More arithmetic width than a small embedded design may require.
11-bit LUT address	Only 2048 sine samples are stored.	Phase truncation produces deterministic spurs.
Q1.31 amplitude	High amplitude precision and direct fixed-point interpretation.	Requires 32-bit storage per table entry.
Full-cycle LUT	Simple direct indexing with no quadrant logic.	Four times the memory of a quarter-wave table.
Sample-by-sample loops	Closely represents a hardware datapath.	Slower than vectorized MATLAB for very long records.

7.2 Practical extensions

- Replace the full-cycle LUT with quarter-wave symmetry to reduce memory while preserving resolution.
- Add phase dithering before truncation to decorrelate and reduce discrete spurious components.
- Introduce an amplitude multiplier or phase offset input for modulation experiments.
- Use integer-only output paths when targeting an FPGA or microcontroller implementation.
- Measure SFDR with coherent sampling and a carefully selected FFT length.

7.3 Validation status

The arithmetic was reproduced independently with unsigned 32-bit phase wrapping, integer tuning words, 11-bit LUT addressing and Q1.31 amplitudes. The measured tuning words, actual frequencies, spectra and LFM time-frequency ridge are internally consistent with the DDS equations. The code also includes input checks and a toolbox-independent STFT to improve portability.

8. Conclusions and delivered files

The practice demonstrates that a DDS can generate accurate fixed and swept-frequency signals with a compact and deterministic digital structure. The 32-bit accumulator provides a frequency resolution of approximately 11.18 microhertz, while the 11-bit LUT address limits waveform phase resolution to 2048 positions per cycle. The resulting tones match the requested frequencies to within a few microhertz, and the LFM spectrogram clearly shows two 500 Hz to 3 kHz linear sweeps.

8.1 Main conclusions

- The tuning word provides direct, predictable control of output frequency.
- Accumulator overflow implements phase wrapping without an explicit reset.
- Phase truncation is the principal memory-versus-spectral-purity trade-off in this design.
- A sample-by-sample tuning-word sequence extends the same DDS core to LFM generation.
- The modular code is suitable for later extensions such as phase offsets, amplitude control, dithering or FPGA-oriented fixed-point verification.

8.2 Package contents

- DDS_Practice_Report.pdf - this English technical report.
- matlab/DDS_main.m - recommended entry point.
- matlab/DDS_single_script.m - self-contained alternative.
- matlab/dds_build_lut.m, dds_generate_tone.m, dds_generate_lfm.m and dds_simple_spectrogram.m - reusable functions.
- matlab/DDS_reference_results.mat - reference parameters and numerical results.
- figures/*.png - reference time-domain, spectrum and spectrogram plots.
- README.txt - execution and package notes.

8.3 Running the practice

```
cd <unzipped-folder>/matlab
DDS_main
```

MATLAB creates a generated_figures directory and overwrites DDS_reference_results.mat with locally generated results. The self-contained file can be used when a single script is preferred; its final spectrogram command requires the Signal Processing Toolbox, whereas the modular version does not.